```
import pandas as pd
```

```
df=pd.read_csv('/content/gender_submission.csv')
```

```
df.isnull().sum()#imform about null and missing values
```

	0
PassengerId	0
Survived	0

dtype: int64

```
#identify Categoriacal Column(Important)
```

```
from sklearn.preprocessing import LabelEncoder
encoder=LabelEncoder()
for col in df.select_dtypes(include='object').columns:
  data[col]=encoder.fit_transform(data[col])
```

```
x=df.iloc[:,0]
y=df.iloc[:,1]
```

```
import numpy as np
x=np.array(x).reshape(-1,1)
y=np.array(y).reshape(-1,1)
print("x array:",x)
print("y array:",y)
```

```
 [1157]
 [1158]
 [1159]
 [1160]
 [1161]
 [1162]
 [1163]
 [1164]
 [1165]
 [1166]
 [1167]
 [1168]
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 [1170]
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 [1174]
 [1175]
 [1176]
 [1177]
 [1178]
 [1179]
 [1180]
 [1181]
 [1182]
 [1183]
 [1184]
 [1185]
 [1186]
 [1187]
 [1188]
 [1189]
 [1190]
 [1191]
 [1192]
 [1193]
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_stat
```

```
x_train
```

```
array([[1228],
       [ 923],
       [ 976],
       [1179],
       [1209],
       [1103],
       [ 986],
       [1281],
       [1219],
       [ 897],
       [ 937],
       [1032],
       [1037],
       [1295],
```

```
       [1250],
       [ 908],
       [1278],
       [1016],
       [1217],
       [ 895],
       [ 910],
       [1024],
       [ 952],
       [1090],
       [1224],
       [ 955],
       [1157],
       [1085],
       [1084],
       [1158],
       [1088],
       [1008],
       [ 918],
       [ 899],
       [1142],
       [1138],
       [1000],
       [ 993],
       [1202],
       [ 921],
       [1087],
       [1282],
       [1010],
       [1163],
       [1006],
       [1203],
       [1023],
       [1049],
       [1044],
       [1263],
       [1291],
       [1164],
       [1292],
       [1272],
       [1001],
       [1068],
       [1258],
       [1299],
```

```
from sklearn.linear_model import LogisticRegression
```

```
model=LogisticRegression(class_weight='balanced')
model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: Dat
  y = column_or_1d(y, warn=True)
```

```
▾ LogisticRegression
LogisticRegression(class_weight='balanced')
```

```
model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: Dat
  y = column_or_1d(y, warn=True)
```

```
▾ LogisticRegression
LogisticRegression(class_weight='balanced')
```

```
y=y_pred=model.predict(x_test)
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score,
```

```
accuracy =accuracy_score(y_test,y_pred)
print("Accuracy:",accuracy)
```

```
Accuracy: 0.5238095238095238
```

```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
print("Confussion Matrix:")
print("TP  FP")
print("FN  TN")
print(cm)
```

```
Confussion Matrix:
TP  FP
FN  TN
[[18 16]
 [24 26]]
```

```
precision=precision_score(y_test,y_pred,zero_division=0)
print("Precision:",precision)
```

```
Precision: 0.42857142857142855
```

```
recall=recall_score(y_test,y_pred,zero_division=0)
print("Recall:",recall)
```

```
Recall: 0.5294117647058824
```

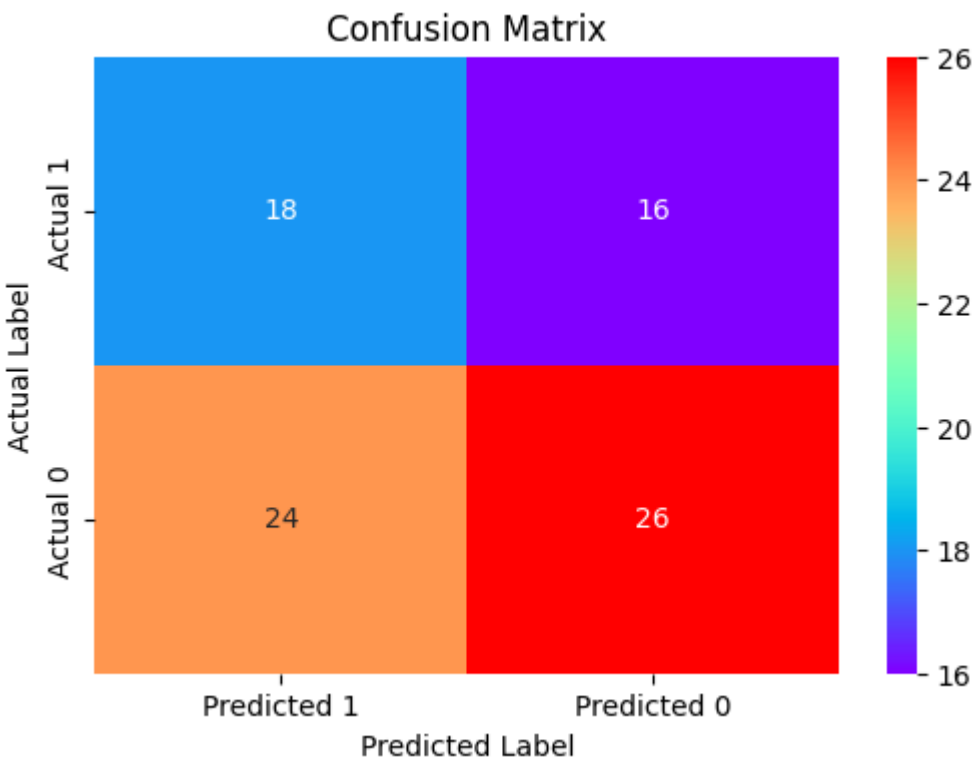
```
f1=f1_score(y_test,y_pred,zero_division=0)
print("F1_score:",f1)
```

```
F1_score: 0.47368421052631576
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='rainbow',
            xticklabels=['Predicted 1','Predicted 0'],
            yticklabels=['Actual 1','Actual 0'])
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.title('Confusion Matrix')
plt.show()
```



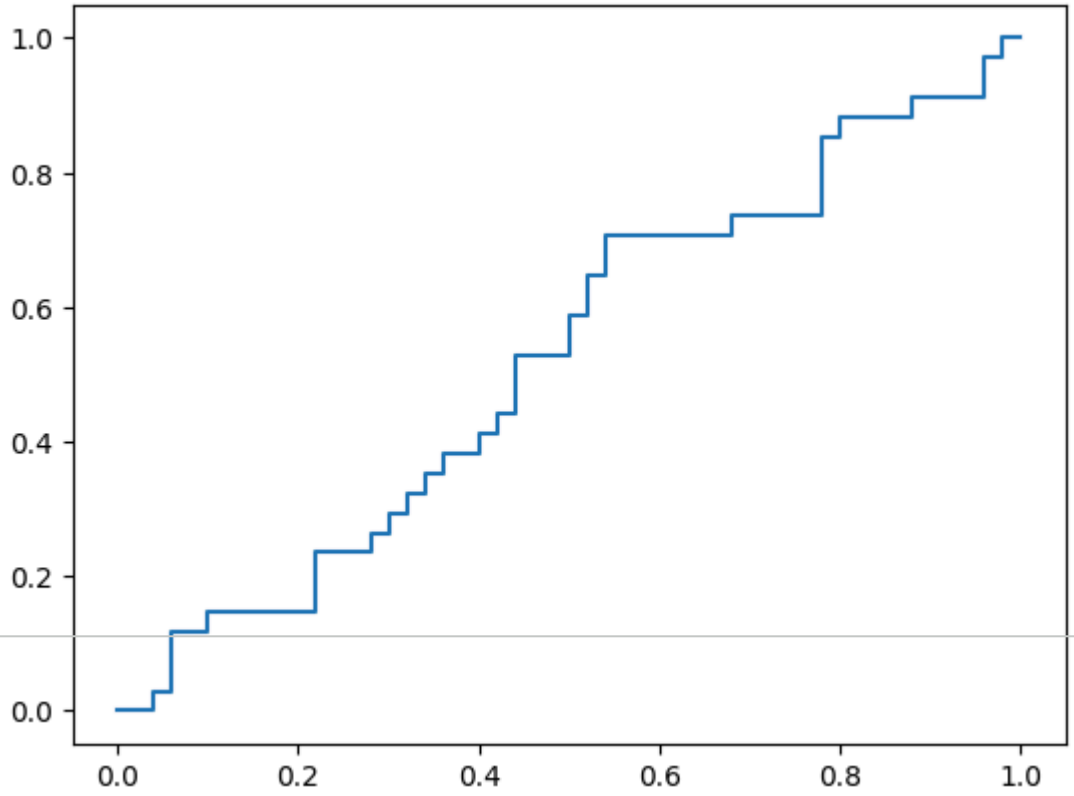

```
from sklearn.metrics import roc_curve,auc
```

```
y_prob=model.predict_proba(x_test)[:,1]
```

```
fpr,tpr,thresholds=roc_curve(y_test,y_prob)
```

```
roc_auc=auc(fpr,tpr)
print('AUC:',roc_auc)
```

```
AUC: 0.5229411764705882
```

```
plt.figure()
plt.plot(fpr,tpr,label="ROC Curve ")
```

```
[<matplotlib.lines.Line2D at 0x7ddf82d433b0>]
```